COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 5

INTEGER DIVISIONS

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THE INTEGERS AND DIVISION

- If *a* and *b* are integers with $a \neq 0$, we say that *a divides b* if there is an integer *c* such that $b = ac$. When *a* divides *b* we say that
  - *a* is a *factor* of *b*
  - *b* is a *multiple* of *a*
- The notation a | b denotes that a divides b
- We write $a \nmid b$ when a does not divide b

Q. Determine whether 3|7 and whether 3|12.

Solution:

- It follows that $3 \nmid 7$, because 7/3 is not an integer.
- On the other hand, 3|12 because 12/3 = 4

Examples

- 77 | 7: false bigger number can't divide smaller positive number
- 7 | 77: true because 77 = 7 x 11
- 24 | 24: true because 24 = 24 x 1
- 0 | 24: false, only 0 is divisible by 0
- 24 | 0: true, 0 is divisible by every number

SOME BASIC PROPERTIES OF DIVISIBILITY OF INTEGERS

- **Theorem**: Let a, b, c be integers. Then
  (i) if a|b and a|c, then a|(b+c)
  (ii) if a|b, then a|bc for all integers c
  (iii) if a|b and b|c, then a|c
  (iv) If a, b, and c are integers such that a|b and a|c, then a|mb+nc whenever m and n are integers.

- For example:

  If 17|34 and 17|170, then 17|204 (34+170 = 204)

  If 17|34, then 17|340 (34 x 10) = 340)

  If 6|12 and 12|144, then 6 | 144

CONGRUENCE

- If a and b are integers and m is a positive integer, then a is congruent to b modulo m if m divides $a - b$.
- We use the notation $a \equiv b \pmod{m}$ to indicate that a is congruent to b modulo m.
- If a and b are not congruent modulo m, we write $a \not\equiv b \pmod{m}$
- **Theorem 3**: Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ iff $a \bmod m = b \bmod m$

1) $3 \equiv 3 \pmod{17}$ True. Any number is congruent to itself (3-3 = 0, divisible by all)

2) $3 \equiv -3 \pmod{17}$ False. (3-(-3)) = 6 isn't divisible by 17.

3) $172 \equiv 177 \pmod{5}$ True. 172-177 = -5 is a multiple of 5

4) $-13 \equiv 13 \pmod{26}$ True. -13 -13 = -26 divisible by 26.

Question: List five integers that are congruent to 3 modulo 7

Answer: 3, 10, 17, 24, 31

CRYPTOLOGY

- Cryptology is the study of secrete messages.

❑ Caesar Cipher

- Mathematical expression:

$$f(p) = (p + 3) \textbf{ mod } 26 \qquad 0 \le p \le 25$$

- Making messages secrets by shifting each letter three letters forward in the alphabet
  - B → E
  - X → A

LETTER ←→ NUMBER CONVERSION TABLE

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12

N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

CAESAR CIPHER

Q. What is the secret message produced from the message "**MEET YOU IN THE PARK**" using the Caesar cipher?

- First replace the letters in the message with numbers. This produces

12 4 4 19 24 14 20 8 13 19 7 4 15 0 17 10

MEET = 12 4 4 19
YOU = 24 14 20
IN = 8 1 3
THE = 19 7 4
PARK = 15 0 17 10

- Now replace each of these numbers p by $f(p) = (p + 3)$ **mod** 26. This produces

15 7 7 22 1 17 23 11 16 22 10 7 18 3 20 13

- Translating this back to letters produces the **encrypted message**

"PHHW BRX LQ WKH SDUN"

CAESAR CIPHER

- To recover the original message from a secret message encrypted by the Caesar cipher, the function f^{-1}, the inverse of f, is used.
- The function f^{-1} sends an integer p from { 0, 1, 2, …25} to

  $f^{-1}(p) = (p - 3)$ **mod** 26

  - In other words, to find the original message, each letter is shifted back three letters in the alphabet, with the first three letters sent to the last three letters of the alphabet.
- The process of determining the original message from the encrypted message is called ***decryption***.

Q: Decrypt the encrypted message **"PHHW BRX LQ WKH SDUN"** to the original message (do it yourself)

SHIFT CIPHER : GENERALIZATION OF CAESAR CIPHER

- There are various ways to generalize the Caesar cipher. For example, instead of shifting each letter by 3, we can shift each letter by k, so that

  $f(p) = (p + k)$ **mod** 26 Such a cipher is called a **shift cipher**.

- The decryption can be carried out using

  $f^{-1}(p) = (p - k)$ **mod** 26

Q. Encrypt the message "**DO NOT PASS GO**" by translating the letters into numbers, applying the encryption function given, and then translating the numbers back into letters.

$f(p) = (p + 13)$ **mod** 26

- The message in number is: 3-14 13-14-19 15-0-18-18 6-14
- Replacing each of these numbers p by $f(p) = (p + 13)$ **mod** 26 produces:

  16-1 0-1-6 2-13-5-5 19-1

- Translating this back into letter produces the encrypted message: "**QB ABG CNFF TB**"

SHIFT CIPHER : GENERALIZATION OF CAESAR CIPHER

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Translating this back into letter produces the encrypted message: "**QB ABG CNFF TB**"

GCD & LCM

Greatest Common Divisor (gcd)

- Let a and b be integers, not both zero. The largest integer d such that $d|a$ and $d|b$ is called the *greatest common divisor* of a and b.
- The *greatest common divisor* of a and b is denoted by gcd(a, b)

Q. What is the greatest common divisor of 24 and 36?

Solution: The positive common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12.
Hence gcd (24,36) = 12

Least Common Multiple (lcm)

- The *least common multiple* of positive integers a and b is the smallest positive integer that is divisible by both a and b.
- lcm (24,36) = 72 [24x3=72 & 36x2=72]

Theorem: Let a and b be positive integers. Then ab = gcd(a,b) x lcm(a,b)

FINDING GCD & LCM USING PRIME FACTORIZATION

- Another way to find the greatest common divisor of two integers is to use the prime factorization of these integers.
- Let prime factorization of the integers a and b, neither equal to zero, are

$$a = p_1^{a1} p_2^{a2} \ldots\ldots p_n^{a_n} ,$$

$$b = p_1^{b1} p_2^{b2} \ldots\ldots p_n^{b_n}$$

$$\mathbf{gcd}(a,b) = p_1^{\min(a1,b1)} \; p_2^{\min(a2,b2)} \ldots\ldots\ldots p_n^{\min(a_n, b_n)}$$

- **Note:** Prime factorization can also be used to find the least common multiple(lcm) of two integers

$$\mathbf{lcm}(a,b) = p_1^{\max(a1,b1)} \; p_2^{\max(a2,b2)} \quad \ldots\ldots\ldots p_n^{\max(a_n, b_n)}$$

FINDING GCD & LCM USING PRIME FACTORIZATION

Q. What is the gcd and lcm of 120 and 500?

Solution: Because the prime factorization of 120 and 500 are $120 = 2^3.3.5$ and $500 = 2^2.5^3$, the greatest common divisor is

$\gcd(120,500) = 2^{\min(3,2)}\ 3^{\min(1,0)}\ 5^{\min(1,3)} = 2^2\ 3^0\ 5^1 = 4 \times 1 \times 5 = 20$

[120/2=60/2=30/2=15/3=5]

$\text{lcm}\ (120,500) = 2^{\max(3,2)}\ 3^{\max(1,0)}\ 5^{\max(1,3)} = 2^3\ 3^1\ 5^3 = 8 \times 3 \times 125 = 3000$

[500/2=250/2=125/5=25/5=5] [Note: $500 = 2^2.\ 3^0.5^3$] where $3^0 = 1$

Q. Find the gcd and lcm(120,500), and then prove the theorem

$ab = \gcd(a,b)\ .\ \text{lcm}(a,b)$, where $a = 120$, and $b = 500$

=> (120 x 500) = (20 x 3000)
=> 60,000 = 60,000

FLOOR & CEILING FUNCTIONS

Q. In asynchronous transfer mode (ATM), data are organized into cells of 53 bytes. **How many ATM cells** can be transmitted in 1 minute over a connection that transmits data at the rate of 500 kilobits per second?

Solution:

- In 1 minute, this connection can transmit 500 x1024 x 60 bits = 30,720,000 bits/second
- Each ATM cell is 53 bytes long, which means that it is 53 x 8 = 424 bits long
- Number of ATM cells that can be transmitted is

  $= \lfloor 30{,}720{,}000/424 \rfloor$

  $= \lfloor 72{,}452.8 \rfloor$

  $= 72{,}452$

REFERENCES

- *Discrete Mathematics and its applications with combinatorics and graph theory (7th edition)* by Kenneth H. Rosen [Indian Adaptation by KAMALA KRITHIVASAN], published by McGraw-Hill, is the required text.
- *SCHAUM'S ouTlines Discrete Mathematics(2nd edition)*, by Seymour Lipschutz, Marc Lipson
- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley